

Szymon Antoni Kubica

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Education

Wrocław University of Science and Technology

- **M.Sc. in Biomedical Engineering** (Medical Informatics) Feb 2025 – Oct 2026
Faculty of Fundamental Problems of Technology
Thesis: *Development of a video-EEG monitoring system prototype*
Supervisor: Assoc. Prof. Cezary Sielużycki, Ph.D., D.Sc., Eng.
Status: in progress, defense planned July 2026
- **B.Sc. in Cybersecurity** (Data Security) Oct 2021 – Jan 2025
Faculty of Information and Communication Technology
Thesis: *Implementation of an example solution based on Extended Berkeley Packet Filter (eBPF) for implementing Moving Target Defense techniques*
Supervisor: Jakub Radtke, M.Sc. Eng.
Diploma grade: 4.5

Research Interests

Operating systems, embedded systems, multimodal biomedical signal analysis (EEG, video, audio), neuroscience, artificial intelligence and machine learning.

Research Experience

Video-EEG monitoring system prototype 2025–2026

Wrocław University of Science and Technology, M.Sc. thesis supervised by Assoc. Prof. C. Sielużycki

- Designed and built a wireless video-EEG prototype enabling synchronized acquisition of EEG signal and camera video.
- Implemented signal acquisition and precise synchronization of video and EEG streams using Lab Streaming Layer (LSL) and custom DSP algorithms in C++ and Python.
- Conducted clinical tests with patients — EEG electrode setup and system validation under real-world conditions.

Conference Activity

- **SECON 2026** — Conference on Electronics, Telecommunications and Energy for Students and Young Researchers (submitted paper)
Towards Edge AI integration in long-term video-EEG monitoring systems for epilepsy — a review of methods and implementation challenges.
- **CYBER 2026** — Doctoral Conference for Young Biomedical Engineering Researchers (submitted paper)
Shared-clock synchronization of independent EEG and video streams for a wireless outpatient video-EEG system.
- Passive participation: HealthTech Synergy Hub Congress, Wrocław AI Symposium, code::dive 2025.

Professional Experience

Software Engineer | SrcCode / KSP

May 2023 – present

- Developing data acquisition applications for measurement devices in C++ (including medical devices).
- Designing and implementing communication protocols between desktop applications and embedded devices (Serial/USB).
- Integrating C/C++ components with Python and working with embedded systems (STM32).

Skills

- **Programming languages:** C++, Python, C, C#, MATLAB, R, SQL
- **Tools:** Linux, CMake, Git, Docker, GDB, Valgrind
- **Languages:** English B2, Polish — native

Additional Qualifications

- IT technician diploma.
- Completed courses: *Beginning C++ Programming*, *Master the Fourier Transform and Its Applications*, *CMake*, *Tests and Tooling for C/C++ Projects*, *Learn Multithreading with Modern C++*, *Qt6* and *QML Intermediate: Interfacing to C++*.
- Participation in clinical research — EEG electrode setup on patients and ERP measurements.